

Review retrospective:
*the anharmonic regularity instance, and an import the build
kept*

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Abstract

A soundness review of `Laplace/OneD/AnharmonicGibbsRegularity.lean`: the `Threepoint.GibbsRegularity` witness for the anharmonic Gibbs measure (the regularity data the FDT review consumes), with its integrability, positivity, and the perturbed pointwise bound/derivative lemmas. Result: **a clean fidelity pass with one docstring fix** — a limit-vs-asymptotic notation correction. A candidate dead import was *rejected* by the build, and a suspected stale cross-reference turned out correct.

1 Setting

This is the regularity factory at the base of the anharmonic FDT/cumulant stack: it proves $\int e^{-tL} < \infty$, $Z > 0$, the perturbed pointwise bound (via AM–GM and coercivity), the pointwise `HasDerivAt`, and assembles them into the FDT review’s `anharmonic_id.gibbsRegularity`.

2 What was reviewed

Six theorems plus the `GibbsRegularity` witness: integrability of e^{-tL} and $x e^{-tL}$, $Z > 0$, the perturbed pointwise `HasDerivAt` and bound, the private bound-integrability, and the headline instance.

3 Fidelity / soundness findings

Sound; no defects. The integrability and positivity are the standard coercive-domination arguments; the perturbed pointwise bound is correct, with the $e^{t/(2c)}$ factor arising from $-thx \leq t|x| \leq \frac{tc}{2}x^2 + \frac{t}{2c} \leq \frac{t}{2}L + \frac{t}{2c}$ (AM–GM plus coercivity $cx^2 \leq L$, $|h| < 1$); the pointwise `HasDerivAt` is the chain rule; and the `GibbsRegularity` assembly feeds the dominated-differentiation theorem the right package. The reviewer confirmed all of this.

One docstring fix. The module doc wrote “ $\partial_h \text{Cov}_h(x, x)|_0 \rightarrow \alpha/(\lambda^3 t)$ as $t \rightarrow \infty$ ” — but the right-hand side is t -dependent (it tends to 0), so “ $\rightarrow$ ” (a finite limit) is the wrong relation. Corrected to the asymptotic “ $\sim \alpha/(\lambda^3 t)$ ” (equivalently $t \partial_h \text{Cov} \rightarrow \alpha/\lambda^3$); the value is right (the cited empirical asymptote 0.005 is $\alpha/(\lambda^3 t)$ at $(\lambda, \alpha, \gamma, t) = (1, 0.5, 1, 100)$).

4 Simplifications applied

Only the docstring fix — the candidate import removal was rejected. `import Laplace.OneD.AnharmonicKappa3Affine` is referenced by no symbol (only in comments, which note a `kappa3` lemma is “private there and re-derived here”), so it looked dead. But removing it makes `integrable_abs_pow_mul_exp_neg_mul_sq` an unknown identifier — that import is the *only* transitive path to the Mathlib Gaussian lemma here. Kept. `open MeasureTheory/Topology` and the other three imports are load-bearing.

5 Disagreements and open questions

None standing. Two reviewer items were resolved against the reviewer by checking: the import is load-bearing (build), and the “AnharmonicKappa3-vs-AnharmonicKappa3Affine” cross-reference the reviewer thought stale is in fact correct — `kappa3.anharmonic_id_id_id_asymptotic` genuinely lives in `AnharmonicKappa3.lean` (a distinct file from the imported `Affine` one, which serves the integrability lemma).

6 Lessons

- **The same transitive lemma, opposite verdicts in adjacent files.** `integrable_abs_pow_mul_exp_neg_mul_sq` was the symbol behind a removable import in the observable-monomials review (reachable elsewhere) but behind a *kept* one here (the sole path). “Is this import dead?” is a per-file build question even for the identical underlying lemma — never carry the verdict across files.
- **Check a “stale cross-reference” before fixing it.** The docstring named `AnharmonicKappa3` while the file imports `AnharmonicKappa3Affine`; the reviewer read that as drift, but the named lemma really is in the non-`Affine` file. A one-line `grep` distinguished a correct cross-file pointer from a typo.
- **“ $\rightarrow$ ” demands a constant right-hand side.** An asymptote with a surviving t is an equivalence ($\sim$) or an $o(\cdot)$ statement, not a limit; the docstring’s “ $\rightarrow \alpha/(\lambda^3 t)$ ” silently asserted a limit that is really 0.

7 Follow-ups

- The remaining anharmonic file `AnharmonicKappa3Affine` and the harmonic regularity/observable/perturbed/affine- κ_3 files; and `IntegralRemainder` (2668L).
- Proof-golf forward tide: the `simp/simpa/change` shortenings the reviewer flagged in the `GibbsRegularity` assembly and the perturbed bound.